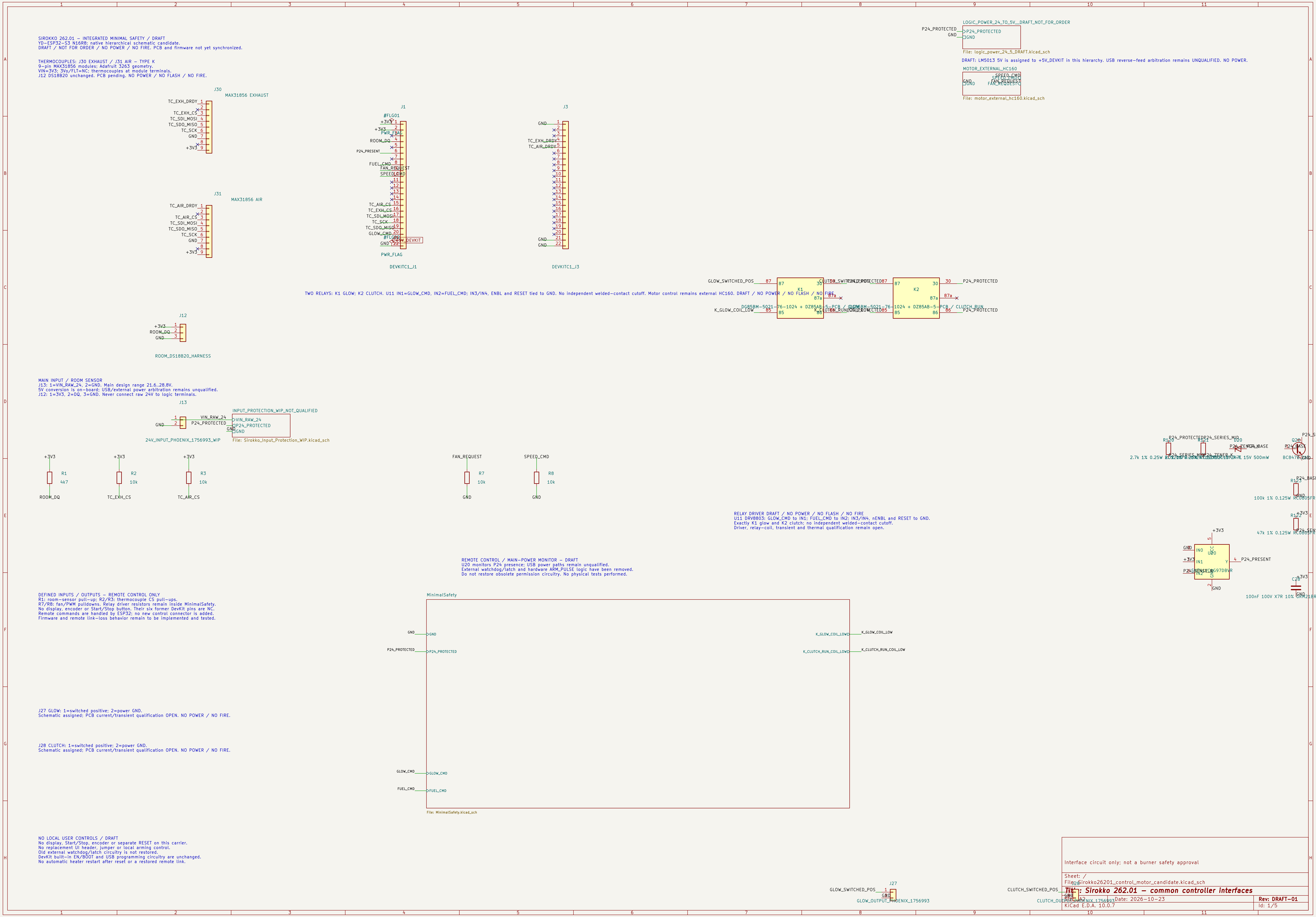


SIROKKO 262.01 – INTEGRATED MINIMAL SAFETY / DRAFT
YD–ESP32–S3 N16R8; native hierarchical schematic candidate.
DRAFT / NOT FOR ORDER / NO POWER / NO FIRE. PCB and firmware not yet synchronized.

THERMOCOUPLES: J30 EXHAUST / J31 AIR – TYPE K
9–pin MAX31856 modules; Adafruit 3263 geometry.
VIN=3V3; 3Vo/FLT=NC; thermocouples at module terminals.
J12 DS18B20 unchanged. PCB pending. NO POWER / NO FLASH / NO FIRE.

DRAFT: LM5013 5V is assigned to +5V_DEVKIT in this hierarchy. USB reverse–feed arbitration remains UNQUALIFIED. NO POWER.

TWO RELAYS: K1 GLOW; K2 CLUTCH. U11 IN1=GLOW_CMD, IN2=FUEL_CMD; IN3/IN4, ENBL and RESET tied to GND. No independent welded–contact cutoff. Motor control remains external HC160. DRAFT / NO POWER / NO FLASH / NO FIRE.

MAIN INPUT / ROOM SENSOR
J13: 1=VIN_RAW_24, 2=GND. Main design range 21.6..28.8V.
5V conversion is on–board; USB/external power arbitration remains unqualified.
J12: 1=3V3, 2=DQ, 3=GND. Never connect raw 24V to logic terminals.

RELAY DRIVER DRAFT / NO POWER / NO FLASH / NO FIRE
U11 DRV8803: GLOW_CMD to IN1; FUEL_CMD to IN2; IN3/IN4, nENBL and RESET to GND.
Exactly K1 glow and K2 clutch; no independent welded–contact cutoff.
Driver, relay–coil, transient and thermal qualification remain open.

REMOTE CONTROL / MAIN–POWER MONITOR – DRAFT
U20 monitors P24 presence; USB power paths remain unqualified.
External watchdog/latch and hardware ARM_PULSE logic have been removed.
Do not restore obsolete permission circuitry. No physical tests performed.

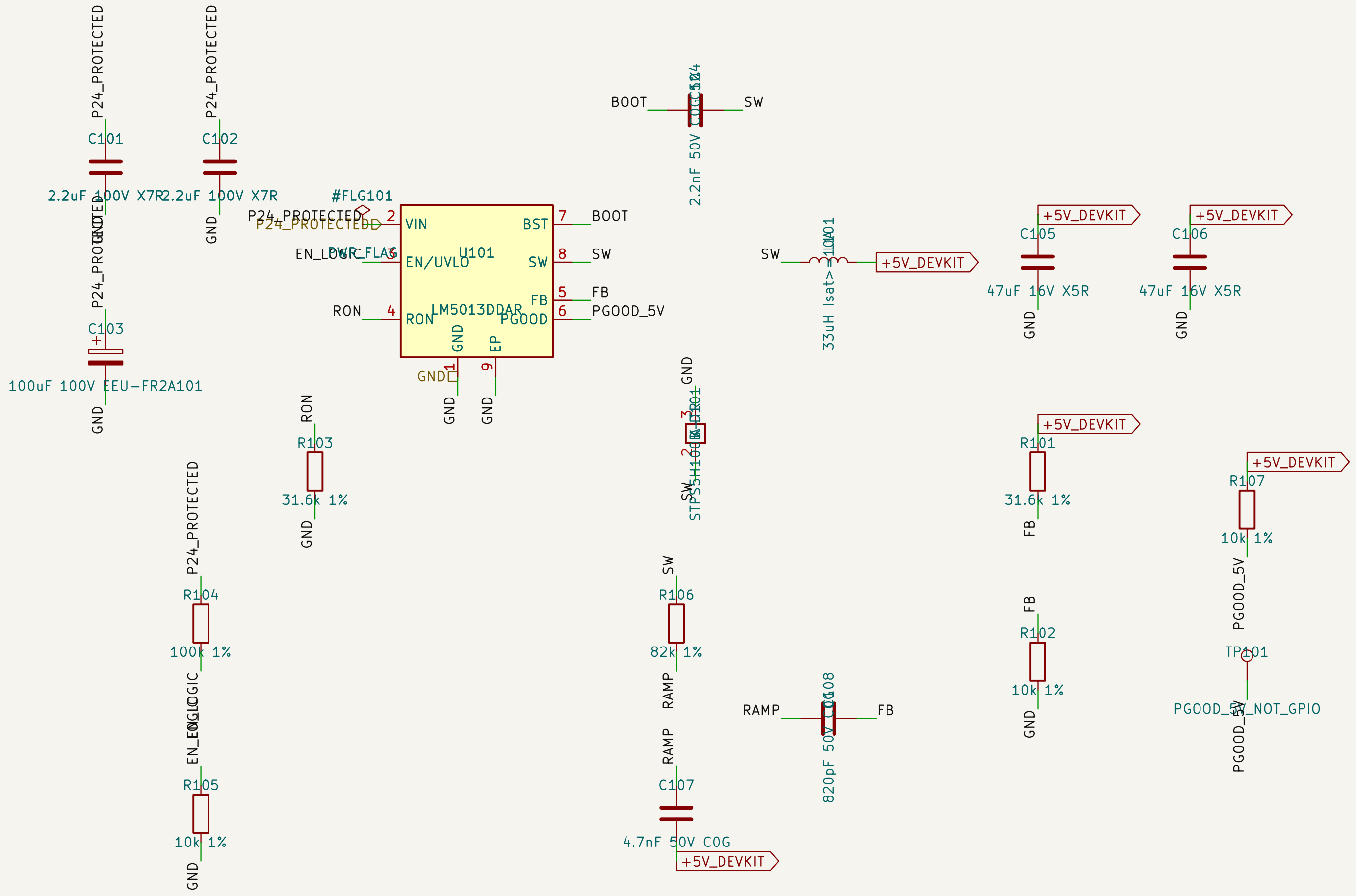
DEFINED INPUTS / OUTPUTS – REMOTE CONTROL ONLY
R1: room–sensor pull–up; R2/R3: thermocouple CS pull–ups.
R7/R8: fan/PWM pulldowns. Relay driver resistors remain inside MinimalSafety.
No display, encoder or Start/Stop button. Their six former DevKit pins are NC.
Remote commands are handled by ESP32; no new control connector is added.
Firmware and remote link–loss behavior remain to be implemented and tested.

J27 GLOW: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

J28 CLUTCH: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

NO LOCAL USER CONTROLS / DRAFT
No display, Start/Stop, encoder or separate RESET on this carrier.
No replacement UI header, jumper or local arming control.
Old external watchdog/latch circuitry is not restored.
DevKit built–in EN/BOOT and USB programming circuitry are unchanged.
No automatic heater restart after reset or a restored remote link.

Interface circuit only; not a burner safety approval
Sheet: /
File: Sirokko26201_control_motor_candidate.kicad_sch
Title: Sirokko 262.01 – common controller interfaces
Size: A2 | Date: 2026–10–23 | **Rev: DRAFT–01**
KiCad E.D.A. 10.0.7 | Id: 1/5



тание логики 24 В → 5 В / 1 А. DRAFT, НЕ ДЛЯ ПРОИЗВОДСТВА.
5013, SNVSCB3 (10.2022), §6-9.2.2.7. Расчётный вход 18-30 В.
о ПОСЛЕ входной защиты; выбросы должны быть ограничены до ≤50 В.
к DevKit и логике на общей PCB. Это не внешний вход 5 В.
онтрольная точка, НЕ GPIO ESP32; не заменяет контроль наличия 24 В.
отключение обратной подпитки USB и силовые каналы здесь ещё не реализованы.
ые места требуют окончательной квалификации; особенно L101 и D101.
щё не выполнена; это отдельный проектный лист, не отдельная внешняя плата.

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KiCad E.D.A. 10.0.7

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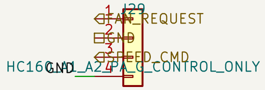

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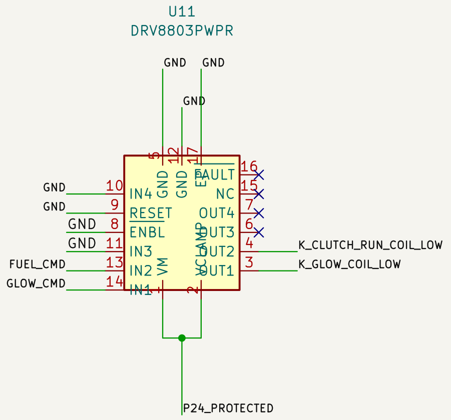
Id: 3/5

SIROKKO: MINIMAL SAFETY CORE – U11 DRV8803 RELAY DRIVER ONLY

DRAFT / NOT FOR ORDER / NO POWER / NO FIRE – hierarchy INTEGRATED in root SCH; PCB/firmware NOT synchronized
CURRENT CONTRACT: external TPS3431 watchdog, SN74HCS74 latch and SN74HCS11 safety logic REMOVED / CANCELLED.

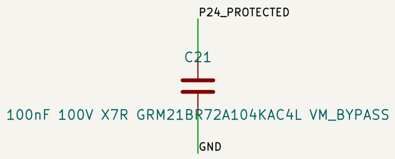
TWO-RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded-contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

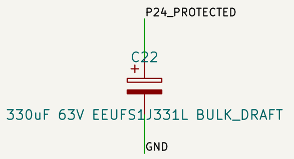
FUTURE FIRMWARE REQUIREMENT (NOT IMPLEMENTED HERE): ESP32-S3 IWDT+TWDT; supervise control task; panic/reboot; keep commands OFF and prohibit automatic restart after reset.

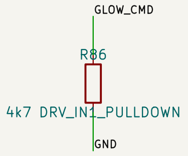


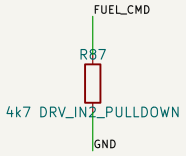
TWO-RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded-contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

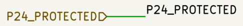
Process checks, timings, purge and software fault handling belong to ESP32; motor/purge power path is unchanged.

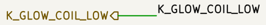


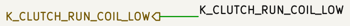


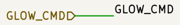












TWO-RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded-contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

24V loss/recovery, startup dynamics, relay ratings and full safety acceptance remain OPEN.

Sheet: /MinimalSafety/
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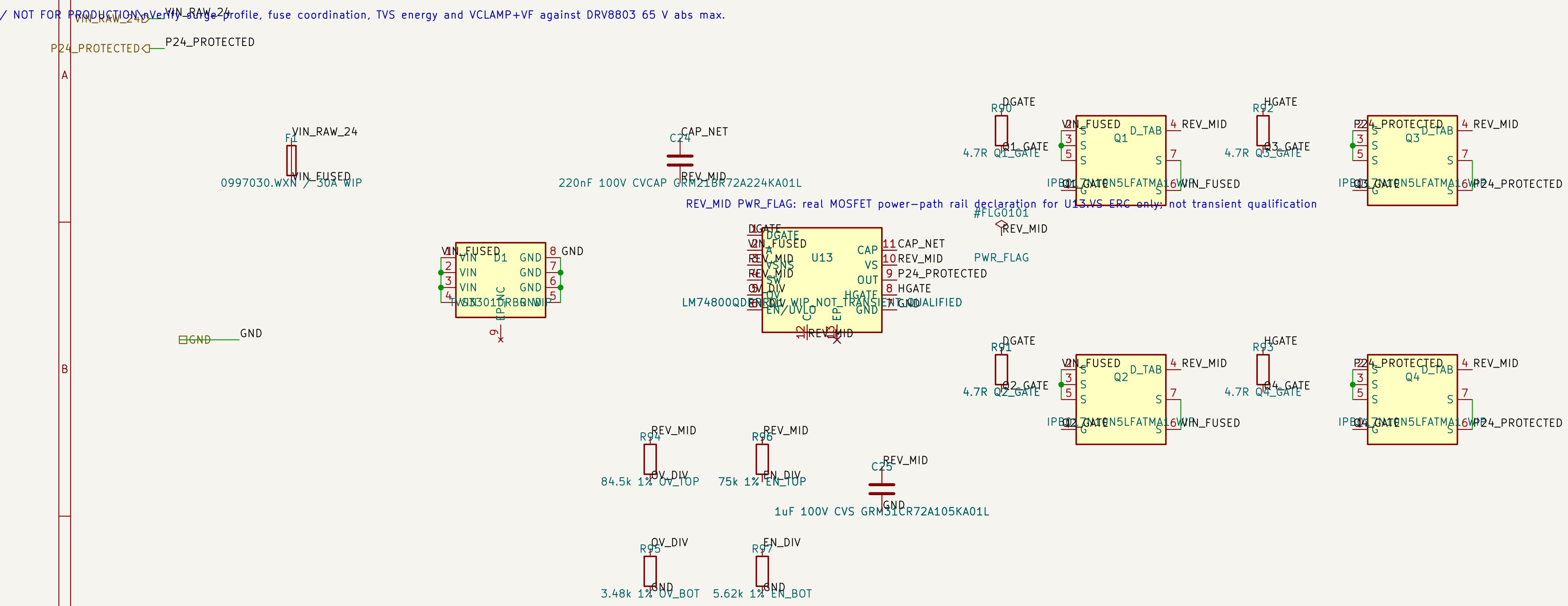
/ NOT FOR PRODUCTION\nVerify surge profile, fuse coordination, TVS energy and VCLAMP+VF against DRV8803 65 V abs max.



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Size: A4 Date:
KiCad E.D.A. 10.0.7

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Id: 7/5